System Request - Study Room Booking System

Project sponsor: Melanie Shell-Weiss, Director of university libraries

Business need: The project has been initiated to improve the process of reserving and managing the library study rooms by providing real-time, self-service booking and usage tracking for students and staff.

Business requirement:

- Online room reservation with real time availability
- Automated check in (QR code) and check out with unclaimed reservation release (within a specified time)
- Notifications and reminder system (text/email) for upcoming and ending reservations
- Usage analytics for room utilization and demand (staff only)

Business value:
We expect to make room reservations more convenient for students and improve the use of available study rooms. It can reduce unused reservations, booking conflicts, and the amount of manual assistance required from library staff. It may also improve student satisfaction by making it easier to find and reserve available study spaces.
Conservative estimates of tangible value to the university per semester include:

- $45000 from staff labor savings
- $30000 from no show bookings (from electricity and other resources)
- Booking disputes reduced by 150 (95%)

Special issue or constraints:

- The Library considers this a strategic system that will modernize its service model and support student expectations for self-service technology.
- Each reservation must have a unique check-in QR code associated with the student's university account.
- Student and reservation data must follow university privacy, retention, and security requirements
- Additional IT services are need to incorporate the SSO with the existing hardwares (doors, locks and card readers)